

FACULTY OF ENGINEERING

Gantry Crane

Engineering Design Report

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Abstract

This report presents the design, development, and testing of a gantry crane system for the transportation of steel metal sheets along a track.

[Replace this placeholder with a 200–350 word summary covering: (1) the problem and its significance, (2) the design approach taken, (3) key results and how they relate to the requirements, and (4) main conclusions and future work recommendations.]

Keywords: Pulley; Gantry Crane; Electro-magnet; Gearbox Design

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Declaration of Originality

I, Frikkie Viljoen, with student number 12345678 declare the following:

This report is my own original work. I further declare that:

1. No part of it has been copied from another person;
2. I did not work with another person on this project and report;
3. I acknowledged all consulted sources in the text and submitted a bibliography;
4. I have indicated all sections where I have used generative AI-enabled technology;
5. Parts without references are my own ideas, arguments, and conclusions.

The Turnitin plagiarism test resulted in a score of Bot Bootlicker%.

_____

Signature

25 November 2026

1 Introduction

This report outlines the design, development, and testing of . . .

[Introduce the broader context of the project. Explain *why* the project is important and what engineering need it addresses. Keep this concise – typically one to two pages.]

The report is structured as follows. Chapter 2 analyses the problem and establishes the project requirements. Chapter 3 presents a literature study of existing solutions. Chapters 4, 5, and 6 cover the concept, preliminary, and detailed design phases respectively. Chapter 7 describes the physical and software implementation. Chapter 8 presents the test results. Chapter 9 draws conclusions and identifies future work.

2 Problem Analysis

2.1 Background

[Describe the domain context and motivation. Explain why the problem exists, what current approaches are insufficient, and why an engineering solution is warranted. Cite relevant sources **example_ref**.]

2.2 Problem Statement

[Clearly state the engineering problem to be solved in one or two focused paragraphs. Avoid proposing solutions here.]

2.3 Project Objectives

The objectives of this project are:

- **Objective 1:** ...
- **Objective 2:** ...
- **Objective 3:** ...

2.4 Requirements

[Requirements should be uniquely identified so they can be traced through design decisions and test results.]

2.4.1 Functional Requirements

- **REQ_01_FUNC:** [The system shall ...]
- **REQ_02_FUNC:** [The system shall ...]
- **REQ_03_FUNC:** [The system shall ...]

2.4.2 Performance Requirements

- **REQ_04_PERF:** [The system shall achieve ...]
- **REQ_05_PERF:** [The system shall achieve ...]

2.4.3 Physical Requirements

- **REQ_06_PHYS:** [The device shall fit within ...]
- **REQ_07_PHYS:** [The device shall weigh less than ...]

2.4.4 Non-Functional Requirements

- **REQ_08_USABILITY:** [An operator with ... training shall be able to ...]
- **REQ_09_MODULARITY:** [The ... subsystem shall be detachable from ...]

2.5 Risk Analysis

[Summarise the risk analysis methodology and the most significant risks identified. A full risk register with likelihood/impact ratings and mitigation strategies is provided in Appendix B.]

3 Literature Study

3.1 Case Studies

[Review published research relevant to your problem. Group related work into subsections. Always cite your sources **example_ref**.]

3.1.1 Case Study 1 – [Topic]

[Discussion of the first body of literature.]

3.1.2 Case Study 2 – [Topic]

[Discussion of the second body of literature.]

3.2 Existing Solutions

[Describe and critically evaluate commercial or academic solutions that already address (partly or fully) the problem. Highlight limitations that motivate your own design.]

3.3 Possible Solutions

[Identify two or three candidate high-level solution strategies. Evaluate each against the requirements established in Chapter 2 and justify the approach selected for further development.]

3.4 Display Data

[If your literature study surfaces specific data (experimental results, performance benchmarks, standards) that will inform the design, present it here with tables or figures as appropriate.]

Figure 3.1. *[Caption describing the figure and its source* **example_ref***].*

Table 3.1. *[Table title* **example_ref***].*

Parameter	**Option A**	**Option B**
Criterion 1	Value	Value
Criterion 2	Value	Value

4 Concept Design

[Describe the high-level solution strategy. Focus on *what* the system must do, not yet *how* it will do it. Use functional flow diagrams and architecture diagrams to illustrate the concept.]

4.1 Functional Flow Diagrams

[Describe the life-cycle and operational flows of the system. Reference each diagram in the text before it appears.]

Figure 4.1. *Functional flow diagram illustrating the system life-cycle.*

Figure 4.2. *Functional flow diagram illustrating the operational flow.*

4.2 System Architecture

[Describe the operational levels of the system (Level 0, Level 1, Level 2, etc.) with a block diagram for each.]

Figure 4.3. *Operational Level 0 architecture.*

Table 4.1. *Operational Functional Units.*

F/U ID	Name	Description
F/U 1.0	[Subsystem A]	[What this functional unit does.]
F/U 2.0	[Subsystem B]	[What this functional unit does.]
F/U 3.0	[Subsystem C]	[What this functional unit does.]

Table 4.2. *Operational Interfaces.*

Interface	From	To	Description
I1	F/U 1.0	F/U 2.0	[Description of data/power/signal.]
I2	F/U 2.0	F/U 3.0	[Description.]

5 Preliminary Design

[Expand the concept into specific subsystem decisions. For each functional unit, identify candidate technologies or components, evaluate them against the requirements, and select and justify the best option.]

5.1 Preliminary Hardware Decisions

5.1.1 [Subsystem 1 – e.g. Imaging Module]

[Discuss the criteria for selecting this component (resolution, power, size, cost, etc.). Present the options in a comparison table.]

Table 5.1. *Available [Subsystem 1] options.*

Component	Criterion A	Criterion B	Criterion C	Selected
Option 1				
Option 2				✓
Option 3				

[Justify the selected option in one or two paragraphs **example_ref**.]

Figure 5.1. *Selected [Subsystem 1] component.*

5.1.2 [Subsystem 2 – e.g. Processing Unit]

Table 5.2. *Single board computer comparison.*

SBC	CPU	RAM	I/O	Selected
Option 1				
Option 2				✓

5.1.3 [Subsystem 3 – e.g. Power Supply]

[Description and justification.]

5.1.4 [Subsystem 4 – e.g. Mechanical Housing]

[Description and justification with sketch or CAD view if available.]

5.2 Preliminary Software Decisions

[Select the operating system, programming language, key libraries, and communication protocols. Justify each choice briefly.]

6 Detailed Design

[Specify every design decision at the component level. Include exact part numbers, electrical parameters, mechanical dimensions, software data structures, and algorithms.]

6.1 Detailed Hardware Decisions

6.1.1 [Component A Specifications]

[Describe the component and why the exact variant was chosen. Refer to the datasheet **example_datash**

Table 6.1. *[Component A] specifications* **example_datasheet**.

Parameter	**Value**
Supply voltage	$5\,\mathrm{V}$
Operating current	$500\,\mathrm{mA}$
Dimensions	$85\,\mathrm{mm} \times 56\,\mathrm{mm}$
Mass	$45\,\mathrm{g}$
Interface	USB 3.0, MIPI CSI-2

6.1.2 [Component B Specifications]

Table 6.2. *[Component B] specifications.*

Parameter	**Value**
Parameter 1	Value
Parameter 2	Value

6.2 Detailed Software Decisions

[Describe the software architecture in detail: threading model, communication protocols, data structures, and key algorithms. Include a software block diagram or state machine where helpful.]

Figure 6.1. *Software architecture diagram.*

6.3 CAD Designs

[Present all custom-fabricated mechanical parts with isometric CAD views and key dimensions annotated.]

6.3.1 [Assembly A – e.g. Probe Enclosure]

[Description of the part, material choice, and manufacturing method.]

Figure 6.2. *Isometric CAD drawing of [Assembly A].*

6.3.2 [Assembly B – e.g. Cable Reel]

Figure 6.3. *Isometric CAD drawing of [Assembly B].*

6.4 Electrical Design

[Present the full electrical schematic or wiring diagram. Describe the power distribution, signal routing, and any protection circuits (fuses, TVS diodes, etc.).]

Figure 6.4. *Full electrical schematic of the system.*

7 Implementation

[Describe how the design was physically realised. This chapter bridges the design (Chapters 5–6) and the testing (Chapter 8).]

7.1 Electrical Implementation

[Describe the PCB or wiring assembly process, any soldering or crimping work, cable management, and final continuity / short-circuit checks performed before power-on.]

Figure 7.1. *Electrical hardware component layout and wiring diagram.*

Figure 7.2. *Photograph of the assembled electrical system.*

7.2 Mechanical Implementation

[Describe the fabrication and assembly of all mechanical parts. Include materials used, tolerances achieved, and any challenges encountered during construction.]

Figure 7.3. *Photograph of the mechanical assembly.*

7.3 Software Implementation

[Describe how the software was written, structured, and tested informally during development. Highlight any algorithms, third-party libraries used, or non-trivial design patterns.]

```
import sensor_library as sl

def init_sensor(pin: int, baud: int = 9600) -> sl.Sensor:
    """Initialise␣and␣calibrate␣the␣sensor␣on␣the␣given␣GPIO␣pin."""
    sensor = sl.Sensor(pin=pin, baud=baud)
    sensor.calibrate()
    return sensor
```

Listing 7.1: Sensor initialisation routine.

7.4 Final System

[Present photographs of the completed system from multiple angles. Briefly confirm that it matches the design intent before formal testing begins.]

(a) *Front view.* (b) *Side view.*

Figure 7.4. *Final assembled system.*

8 Testing

[Each test traces back to a requirement defined in Section 2.4. For every test, state the objective, procedure, results, and a pass/fail verdict.]

8.1 Test REQ_04_PERF – [Requirement Short Name]

Objective: Verify that the system achieves . . . as specified in REQ_04_PERF.

Procedure:

1. Set up the system in the test configuration.

Step 2 . . .

2. Record the measured value.
3. Repeat three times and calculate the average.

Acceptance criterion: The measured value shall be $\leq 15\,\mathrm{ms}$.

Results:

Table 8.1. *Results for Test REQ_04_PERF.*

Trial	Expected (ms)	Measured (ms)	Pass / Fail
1	$\leq 15\,\mathrm{ms}$		
2	$\leq 15\,\mathrm{ms}$		
3	$\leq 15\,\mathrm{ms}$		
Average			

Conclusion: Requirement REQ_04_PERF was [**met** / **not met**] because . . .

8.2 Test REQ_05_PERF – [Requirement Short Name]

Objective: . . .

Procedure:

1. . . .

Acceptance criterion: . . .

Results:

Table 8.2. *Results for Test REQ_05_PERF.*

Trial	Expected	Measured	Pass / Fail
1			
2			
3			

Conclusion: . . .

9 Conclusion

[Summarise what was achieved relative to the objectives stated in Section 2.3. Address each objective explicitly.]

Objective 1 was [achieved / partially achieved / not achieved] because . . .

Objective 2 was [achieved / partially achieved / not achieved] because . . .

Objective 3 was [achieved / partially achieved / not achieved] because . . .

[Comment on the overall performance of the system during testing. Identify limitations encountered and explain their likely causes.]

Future Work

[List and briefly describe improvements or extensions recommended for future iterations of the project. Common examples include:]

nsor integration

ted control loop

ure for field use

A Generative AI Usage

[List every instance where generative AI tools were used to assist in writing or editing any part of this report, as required by the Declaration of Originality.]

Table A.1. *Generative AI (GA) evidence throughout the report.*

Section	Tool	Prompt summary	How output was used
§7.3	ChatGPT-4o	"Explain threading model for . . . "	Edited, verified, and rewritten in own words.

B Risk Register

Table B.1. *Risk identification and management.*

ID	Risk description	Likelihood	Impact	Mitigation strategy
R01	Component delivery delay	Medium	High	Order components 4 weeks in advance; identify local suppliers.
R02	Waterproofing failure of enclosure	Low	High	Pressure-test housing before electronics are installed.
R03	Software instability on embedded platform	Medium	Medium	Use watchdog timer; implement error logging.
R04	[Add risk]			

Table B.2. *Risk traceability matrix **BEFORE** mitigation.*

	Impact				
Likelihood	**1 – Negligible**	**2 – Minor**	**3 – Moderate**	**4 – Major**	**5 – Severe**
5 – Very high					
4 – High				R01	
3 – Medium			R03		
2 – Low				R02	
1 – Very low					

Table B.3. *Risk traceability matrix **AFTER** mitigation.*

	Impact				
Likelihood	**1 – Negligible**	**2 – Minor**	**3 – Moderate**	**4 – Major**	**5 – Severe**
5 – Very high					
4 – High					
3 – Medium		R01	R02		
2 – Low		R03			
1 – Very low					

C Additional Technical Data

[Include supplementary material that is too detailed for the main body but necessary for completeness or reproducibility. Typical content includes:]

- Full electrical schematics (if too large for Chapter 6)
- Complete test data tables
- Extended derivations or calculations
- Full source-code listings
- Manufacturer datasheets (as PDF pages via `\includepdf`)